



MQTT JSON & Optimization

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WATTIUS BATTERY SOLUTIONS

Camí de Can Bassa, 6-8
08401 Granollers (Barcelona)
SPAIN
info@wattius.com
+34 93 018 06 08

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Table of Contents

1. Topics	2
2. Optimization	3
3. Measurements variables	5
4. JSONs	8
4.1. Measurements (all JSON)	8
4.2. Events	14

1. Topics

Each device has 3 different topics associated. These topics are composed of an initial wBMS structure followed by 4 additional fields:

wBMS/version/SerialNumber/Type/Direction

- version: the current MQTT messages version is v1.
- SerialNumber (SN): serial number of the BMU, it can be obtained by either scanning the QR code of the board or connecting to the wBMS-Toolkit. Note that the SN length must be 12 digits (in hexadecimal), so 0's should be placed before reaching the required length.
- Type: depending on the purpose this field can have two different values:
 - measurements: live data reporting.
 - events: used to send the DTCs.
 - monitoring: used to establish communication with the BMU and the wBMS-Toolkit.
- Direction: depending on whether the messages are published by the wBMS BMU or not this field can have two different values:
 - in: received by the wBMS BMU
 - out: send by the wBMS BMU

Below an example of the topics used for the device with serial number 000541000C2C are shown:

- wBMS/v1/000541000C2C/measurements/out
- wBMS/v1/000541000C2C/events/out
- wBMS/v1/000541000C2C/monitoring/out
- wBMS/v1/000541000C2C/monitoring/in

The customer must set the appropriate policies in the MQTT broker to enable the previous topics. If not, the communication packets will be dropped, and no communication will be established between devices regardless of the BMU settings.

2. Optimization

The measurements topic is used to send internal BMS data in JSON format, with a maximum message size of 15 kB. The BMS only transmits variables that change between samples to reduce cloud costs and avoid sending repeated information.

A sample is taken every 5 seconds. At each sample, the BMS compares all variables with the previous transmitted sample. A variable is marked as changed when the difference exceeds its defined threshold.

Variable	Threshold	Unit
battery_voltage	500	mV
power	100	W
battery_current	500	mA
counters charge	1000	A·s
counters discharge	1000	A·s
SoC	100	0,01 %
SoH	100	0,01 %
max_cell_voltage value	10	mV
min_cell_voltage value	10	mV
voltage c00x	10	mV
max_cell_temperature_value	100	0,01 °C
min_cell_temperature_value	100	0,01 °C
avg_cell_temperature_value	100	0,01 °C
max_BMS_temperature_value	100	0,01 °C
min_BMS_temperature_value	100	0,01 °C
bms_temperature t00x	1000	0,01 °C
avg_cell_temperature_value	10	0,01 °C
avg_cell_voltage_value	10	mV
system_current	500	mA
system_soc	100	0,01 %
system_chargemanager_chargecurrentmax	500	mA
system_chargemanager_chargecurrentmin	500	mA

system_maxchargevoltage	100	mV
system_minchargevoltage	100	mV
reference_voltage	500	mV

A full message is also sent every 120 samples (600 seconds), even if no changes are detected.

Variable changes are only checked at the end of each sampling period. If a variable exceeds the threshold for a moment but returns to its previous value before the sample is taken, the change is not recorded.

Some variables must always be transmitted together due to their structure. This mainly affects the error and event arrays.

With this optimization, most messages are smaller than 15 kB. Instead of sending them immediately, several messages may be accumulated until reaching the size limit. The system also applies a 10 minute maximum waiting time to avoid long periods without communication.

In some configurations with a high number of cells, a single message may exceed the maximum size. In these cases, the BMS splits the content into multiple valid JSON messages.

3. Measurements variables

	Field name	Type	Units	Scope (BP / BS)	Description
HEADER	msg_part	String	-	-	Internal variable for debug purposes.
	name	String	-	Battery Pack	Serial number of the BMU
	hw	String	-	Battery Pack	Hardware version of the board
	sw	String	-	Battery Pack	Main microcontroller firmware version
	gw	String	-	Battery Pack	Gateway microcontroller firmware version
	timestamp	Number	ms	-	Unix timestamp of the moment of wBMS message capture
	systime_sample	Number	-	-	Internal variable for debug purposes.
	systime_delta	Number	-	-	Internal variable for debug purposes.
	configuration_crc	String	-	-	CRC of the current configuration.
BODY	battery_voltage	Number	V	Battery Pack	See "Battery voltage" signal from CAN message 0x560 + BMU ID
	batteryStatus	Object	-	BatteryPack	Object containing information related with BMU status (MainStatus).
	batteryStatus MainStatus	Number	ENUM	Battery Pack	See "BMU status" signal from CAN message 0x510 + BMU ID
	chargeManager	Object	-	Battery Pack	Object containing information related with charge/discharge (Status, ChargeCurrentMax and DischargeCurrentMax).
	chargeManager Status	Number	ENUM	Battery Pack	See "Charge/Discharge control state" signal from CAN message 0x510 + BMU ID
	chargeManager ChargeCurrentMax	Number	A	Battery Pack	See "Charge current setpoint" signal from CAN message 0x520 + BMU ID
	chargeManager DischargeCurrentMax	Number	A	Battery Pack	See "Discharge current setpoint" signal from CAN message 0x520 + BMU ID
	Auxiliary NTC	Object	-	Battery Pack	Object containing information related with BMU external temperatures (NTC_value).
	Auxiliary NTC NTC_value	Number	°C	Battery Pack	See "External temperature 4" signal from CAN message 0x5A0 + BMU ID (3/3)
	HV_sensor	Object	-	Battery Pack	Object containing information related with HV sensor (HV_sensor_Value).
	HV_sensor HV_sensor_Value	Number	mV	Battery Pack	See "Voltage sensor: battery voltage" signal from CAN message 0x5A0 + BMU ID (2/3)
	SoC_Info	Object	-	Battery Pack	Object containing information related with BMU SoC (Local_SoC).
	SoC_Info Local_SoC	Number	%	Battery Pack	See "State of Charge (SoC)" signal from CAN message 0x5C0 + BMU ID
	counters	Object	-	Battery Pack	Object containing information related with maximeters (charge, discharge, total_cycles_equivalent).

counters charge	Number	Ah	Battery Pack	See "Total charged capacity" signal from CAN message 0x5E0 + BMU ID(4/5)
counters discharge	Number	Ah	Battery Pack	See "Total discharged capacity" signal from CAN message 0x5E0 + BMU ID(5/5)
counters total_cycles_equivalent	Number	Cycles	Battery Pack	See "Equivalent cycles" signal from CAN message 0x5E0 + BMU ID(2/5)
SoC	Number	%	Battery Pack	See "State of Charge (SoC)" signal from CAN message 0x5C0 + BMU ID
SoH	Number	%	Battery Pack	See "State of Health (SoH)" signal from CAN message 0x5C0 + BMU ID
power	Number	W	Battery Pack	Power delivered/consumed by the Battery pack.
battery_current	Number	A	Battery Pack	See "Battery current" signal from CAN message 0x540 + BMU ID
max_cell_voltage	Object	-	Battery Pack	Object containing information related with the maximum cell voltage (value, cell_id).
max_cell_voltage value	Number	V	Battery Pack	See "Maximum cell voltage" signal from CAN message 0x580 + BMU ID(1/3)
max_cell_voltage cell_id	String	-	Battery Pack	Identification number of the "Maximum cell voltage". Storing format: "string_ID" + c + "cell_id_inside_the_string" e.g. 3c24 (cell 24 of string 3).
min_cell_voltage	Object	-	Battery Pack	Object containing information related with the minimum cell voltage (value, cell_id).
min_cell_voltage value	Number	V	Battery Pack	See "Minimum cell voltage" signal from CAN message 0x580 + BMU ID(1/3)
min_cell_voltage cell_id	String	-	Battery Pack	Identification number of the "Minimum cell voltage". Storing format: "string_ID" + c + "cell_id_inside_the_string" e.g. 3c24 (cell 24 measurement from string 3).
avg_cell_voltage_value	Number	V	Battery Pack	See "Cell voltage average" signal from CAN message 0x580 + BMU ID(3/3)
max_cell_temperature_value	Number	°C	Battery Pack	See "Maximum cell temperature" signal from CAN message 0x580 + BMU ID(2/3)
min_cell_temperature_value	Number	°C	Battery Pack	See "Minimum cell temperature" signal from CAN message 0x580 + BMU ID(2/3)
avg_cell_temperature_value	Number	°C	Battery Pack	See "Average cell temperature" signal from CAN message 0x580 + BMU ID(2/3)
max_BMS_temperature_value	Number	°C	Battery Pack	See "Maximum balancing temperature" signal from CAN message 0x580 + BMU ID(3/3)
min_BMS_temperature_value	Number	°C	Battery Pack	See "Minimum balancing temperature" signal from CAN message 0x580 + BMU ID(3/3)
system_current	Number	A	Battery System	See "Actual battery current" signal from CAN message 0x341
system_soc	Number	%	Battery System	See "SoC" signal from CAN message 0x341

system_chargemanager_chargecurrentmax	Number	A	Battery System	See "Max. battery charge current" signal from CAN message 0x241
system_chargemanager_chargecurrentmin	Number	A	Battery System	See "Max. battery discharge current" signal from CAN message 0x241
system_maxchargevoltage	Number	V	Battery System	See "Max. battery voltage" signal from CAN message 0x241
system_minchargevoltage	Number	V	Battery System	See "Min. battery voltage" signal from CAN message 0x241
reference_voltage	Number	V	Battery System	See "Actual battery voltage" signal from CAN message 0x341
voltage1	Object	-	Battery Pack	Object containing all cell voltage measurements (cXXX).
voltage1 c00x	Number	V	Battery Pack	Cell voltage measurement X
cell_temperature1	Object	-	Battery Pack	Object containing all cell temperature measurements (tXXX).
cell_temperature1 t00x	Number	°C	Battery Pack	Cell temperature measurement X
bms_temperature1	Object	-	Battery Pack	Object containing all cell voltage measurements (tXXX).
bms_temperature1 t00x	Number	°C	Battery Pack	Cell balancing temperature X
balancing1	Object	-	Battery Pack	Object containing all cell balancing status (cXXX).
balancing1 c00x	Number	Flag	Battery Pack	Cell balancing status X
error	Array	Flag	Battery Pack	Array containing all error flags See "Errors message" signal from CAN message 0x530
event	Array	Flag	Battery Pack	Array containing all warning flags See "Errors message" signal from CAN message 0x550

4. JSONs

4.1. Measurements (all JSON)

```
{
  "msg_part": String, (e.g. "1")
  "name": String, (e.g. "000541000C2C")
  "hw": String, (e.g. "4.1.0")
  "sw": String, (e.g. "3.7.5")
  "gw": String, (e.g. "1.2.0")
  "timestamp": Number, (e.g. 1748273670000)
  "systime_sample": Number, (e.g. 360369)
  "systime_delta": Number, (e.g. 201)
  "configuration_crc": String, (e.g. "0xB5114AB")
  "battery_voltage": Number, (e.g. 2004.288)
  "batteryStatus": (Object)
  {
    "MainStatus": Number (e.g. 3)
  },
  "chargeManager": (Object)
  {
    "Status": Number, (e.g. 0)
    "ChargeCurrentMax": Number, (e.g. 0)
    "DischargeCurrentMax": Number, (e.g. 0)
  },
  "Auxiliary NTC": (Object)
  {
    "NTC_value": Number, (e.g. -49.67)
  },
  "HV_sensor": (Object)
  {
    "HV_sensor_Value": Number, (e.g. 24000)
  },
  "SoC_Info": (Object)
  {
    "Local_SoC": Number, (e.g. 19.00)
  },
  "counters": (Object)
  {
    "charge": Number, (e.g. 11.94)
```

HEADER

```

"discharge": Number (e.g. 36.77)
"total_cycles_equivalent": Number (e.g. 0)
},
"SoC": Number, (e.g. 19.00)
"SoH": Number, (e.g. 100.00)
"power": Number, (e.g. 0)
"battery_current": Number, (e.g. 0)
"max_cell_voltage": (Object)
{
  "value": Number, (e.g. 3.460)
  "cell_id": String, (e.g. "1c005")
},
"min_cell_voltage": (Object)
{
  "value": Number, (e.g. 3.458)
  "cell_id": String, (e.g. "1c014")
},
"avg_cell_voltage_value": Number, (e.g. 3.479)
"max_cell_temperature_value": Number, (e.g. 28.55)
"min_cell_temperature_value": Number, (e.g. 28.55)
"avg_cell_temperature_value": Number, (e.g. 28.55)
"max_BMS_temperature_value": Number, (e.g. 28.55)
"min_BMS_temperature_value": Number, (e.g. 28.55)
"system_current": Number, (e.g. 0,000)
"system_soc": Number, (e.g. 19.00)
"system_chargemanager_chargecurrentmax": Number, (e.g.0.0)
"system_chargemanager_chargecurrentmin": Number, (e.g.0.0)
"system_maxchargevoltage": Number, (e.g. 2417.4)
"system_minchargevoltage": Number, (e.g. 1440.0)
"reference_voltage": Number, (e.g. 2004.288)
"voltage1": (Object)
{
  "c001": Number, (e.g. 3.459)
  "c002": Number, (e.g. 3.459)
  "c003": Number, (e.g. 3.459)
  ...
  "cXXX": Number, (e.g. 3.459)
},
"cell_temperature1": (Object)
{

```

```

"t001": Number, (e.g. 28.55)
"t002": Number, (e.g. 28.55)
"t003": Number, (e.g. 28.55)
...
"tXXX": Number, (e.g. 28.55)
},
"bms_temperature1": (Object)
{
"t001": Number, (e.g. 28.55)
"t002": Number, (e.g. 28.55)
"t003": Number, (e.g. 28.55)
...
"tXXX": Number, (e.g. 28.55)
},
"balancing1": (Object)
{
"c001": Number, Flag
"c002": Number, Flag
"c003": Number, Flag
...
"cXXX": Number, Flag
},
"error": (Array[64])
[
  Number, Flag (Battery System errors)
  Number, Flag (Architecture configuration errors)
  Number, Flag (Parallelization voltage errors)
  Number, Flag (General configuration errors)
  Number, Flag (Parallelization error: communication lost with primary BMU)
  Number, Flag (Cell overvoltage)
  Number, Flag (Cell undervoltage)
  Number, Flag (Communication error with CMU)
  Number, Flag (Internal CMU error)
  Number, Flag (Internal CMU temperature error)
  Number, Flag (Internal CMU voltage reference error)
  Number, Flag (Battery Pack short circuit)
  Number, Flag (Battery Pack overcurrent)
  Number, Flag (Overload ratio exceeded)
  Number, Flag (Current sensor errors)
  Number, Flag (BMU temperature errors)

```

Number, *Flag* (Cell overtemperature)
Number, *Flag* (Cell undertemperature)
Number, *Flag* (Balancing overtemperature)
Number, *Flag* (Balancing undertemperature)
Number, *Flag* (Maximum operating cell temperature delta)
Number, *Flag* (Thermal runaway)
Number, *Flag* (NTC Advanced Diagnostics: Open-wire or shortcircuit detected on cell)
Number, *Flag* (BMU external temperature overtemperature)
Number, *Flag* (String undervoltage)
Number, *Flag* (BMU supply voltage error)
Number, *Flag* (String overvoltage)
Number, *Flag* (Cells unbalanced)
Number, *Flag* (Advanced Relaxation Voltage Delta error)
Number, *Flag* (Internal memory errors)
Number, *Flag* (Internal BMU error)
Number, *Flag* (Safety sensor error)
Number, *Flag* (High voltage sensor errors)
Number, *Flag* (IMD errors)
Number, *Flag* (Isolation error)
Number, *Flag* (Battery Pack SoC errors)
Number, *Flag* (Battery Pack SoH error)
Number, *Flag* (Contactor stuck open errors)
Number, *Flag* (Contactor stuck closed errors)
Number, *Flag* (Contactor supply error)
Number, *Flag* (Precharge error)
Number, *Flag* (Emergency signal error)
Number, *Flag* (Actuation errors)
Number, *Flag* (Auto OFF)
Number, *Flag* (Mode mismatch)
Number, *Flag* (Open-wire errors)
Number, *Flag* (Output driver errors diagnostics)
Number, *Flag* (User forced errors)
Number, *Flag* (SD error)
Number, *Flag* (Ethernet module error)
Number, *Flag* (Reserved)
Number, *Flag* (Reserved)
Number, *Flag* (Reserved)
Number, *Flag* (Reserved)
Number, *Flag* (Reserved)
Number, *Flag* (Reserved)

```
Number, Flag (Reserved)
Number, Flag (Reserved)
Number, Flag (Reserved)
Number, Flag (Reserved)
Number, Flag (Reserved)
Number, Flag (Reserved)
Number, Flag (Reserved)
]
"event": (Array [64])
[
  Number, Flag (Battery System errors)
  Number, Flag (Architecture configuration errors)
  Number, Flag (Parallelization voltage errors)
  Number, Flag (General configuration errors)
  Number, Flag (Parallelization error: communication lost with primary BMU)
  Number, Flag (Cell overvoltage)
  Number, Flag (Cell undervoltage)
  Number, Flag (Communication error with CMU)
  Number, Flag (Internal CMU error)
  Number, Flag (Internal CMU temperature error)
  Number, Flag (Internal CMU voltage reference error)
  Number, Flag (Battery Pack short circuit)
  Number, Flag (Battery Pack overcurrent)
  Number, Flag (Overload ratio exceeded)
  Number, Flag (Current sensor errors)
  Number, Flag (BMU temperature errors)
  Number, Flag (Cell overtemperature)
  Number, Flag (Cell undertemperature)
  Number, Flag (Balancing overtemperature)
  Number, Flag (Balancing undertemperature)
  Number, Flag (Maximum operating cell temperature delta)
  Number, Flag (Thermal runaway)
  Number, Flag (NTC Advanced Diagnostics: Open-wire or shortcircuit detected on cell)
  Number, Flag (BMU external temperature overtemperature)
  Number, Flag (String undervoltage)
  Number, Flag (BMU supply voltage error)
  Number, Flag (String overvoltage)
  Number, Flag (Cells unbalanced)
  Number, Flag (Advanced Relaxation Voltage Delta error)
  Number, Flag (Internal memory errors)
```

[illegible]

4.2. Events

Structure:

```
{
  "name": String, (e.g. "000541000C2C")
  "timestamp": Number, (e.g. 1748273670000)
  "type": (Object with type name "event" or "flag")
  {
    "descr": String, (e.g. "Error Injected via wBMS-Toolkit"
                  or "[Finished] Error Injected via wBMS-Toolkit")
    "code": Number, (e.g. 10)
  }
}
```

Example:

```
{
  "name": "000541000F76",
  "timestamp": 1765808221007,
  "event":
  {
    "descr": " Battery Pack activation requested (Start signal)",
    "code": 1
  }
},
{
  "name": "000541000F76",
  "timestamp": 1765808224008,
  "event":
  {
    "descr": " Positive Main Power Relay activation requested",
    "code": 147
  }
},
{
  "name": "000541000F76",
  "timestamp": 1765808242009,
  "event":
  {
    "descr": "[Finished] Battery Pack activation requested (Start signal)",
```

```
"code": 1
},
{
  "name": "000541000F76",
  "timestamp": 1765808245010,
  "event":
  {
    "descr": "[Finished] Positive Main Power Relay activation requested",
    "code": 147
  }
},
{
  "name": "000541000F52",
  "timestamp": 1765808794044,
  "flag":
  {
    "descr": " Error Injected via wBMS-Toolkit",
    "code": 89
  }
}
{
  "name": "000541000F52",
  "timestamp": 1765808803046,
  "flag":
  {
    "descr": "[Finished] Error Injected via wBMS-Toolkit",
    "code": 89
  }
}
{
  "name": "000541000F53",
  "timestamp": 1765808805046,
  "flag":
  {
    "descr": "[Warning] CAN bus malfunction",
    "code": 251
  }
}
```

```
{  
  "name": "000541000F54",  
  "timestamp": 1765808805046,  
  "flag":  
  {  
    "descr": "[Finished] CAN bus malfunction",  
    "code": 251  
  }  
}
```